

Unit - II Virtualization

Weightage marks : 16marks

Syllabus

TLO 2.1 Explain features of Virtualization.
TLO 2.2 Compare characteristics of Virtualization types.
TLO 2.3 Write the steps to build a virtual machine using VMWare on the given Operating System.
TLO 2.4 Differentiate Virtual Machine Migration, Consolidation and Management.
TLO 2.5 Explain advantages and disadvantages of Virtualization.

Unit - II Virtualization

2.1 Introduction, Virtualization Reference Model, Characteristics of virtualized environment
2.2 Differentiate various types of Virtualization : Storage, Network , Desktop , Application server
2.3 Technology Examples
2.3.1 VMWare: Full Virtualization Reference Model
2.3.2 Xen: Architecture and Guest Operating System Management
2.4 Definition and Life Cycle of Virtual Machine(VM), VM Migration: Concept and Techniques, VM Consolidation: Concepts, VM Management: Concepts
2.5 Advantages and Disadvantages of Virtualization

Flipped
Classroom
Presentations
Lecture Using
Chalk-Board
Video
Demonstrations

Introduction (What is basically Virtualization)

- **Virtualization** lets one computer act like many by running multiple virtual machines (VMs) on a single physical system. This helps use the full power of the computer, instead of wasting it on one task.
- In **cloud computing**, virtualization allows cloud providers to divide one big server into many small virtual ones, saving hardware and cost.
- In this you will learn what a virtual machine is, why it is important, the different types of virtualization, how it works, and the benefits and disadvantages associated with it.

Virtualization in Cloud Computing

Virtualization is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources".

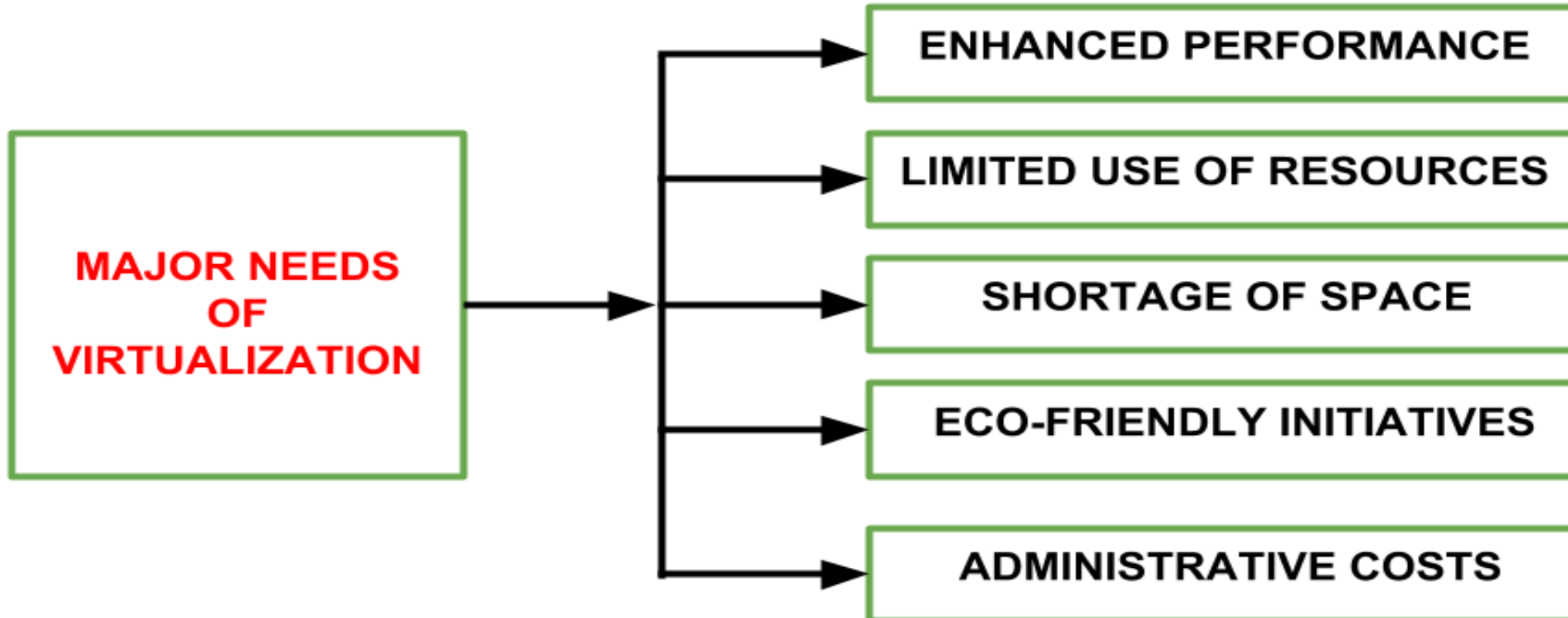
In other words, Virtualization is a technique, which allows to share a single physical instance of a resource or an application among multiple customers and organizations. It does by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.



Example

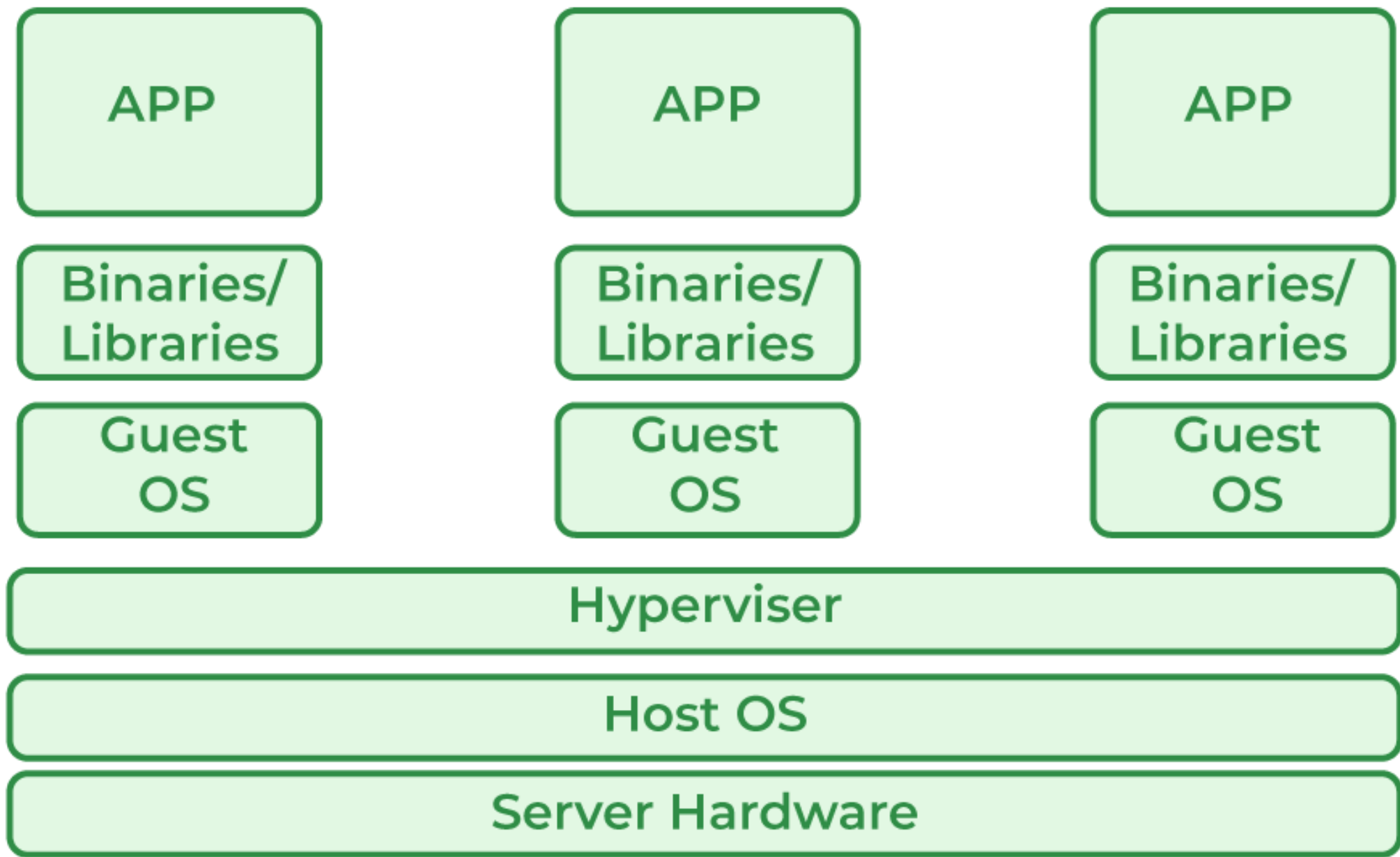
- Think of a big house where only one person lives. Most of the rooms are empty and not being used. Now, if you are dividing the house into smaller rooms and renting each one to different people. Everyone gets their own space to live, cook, and sleep all inside the same house.
- Let's try to understand this example how this house behaves like virtualization:
- The **house** is like one real computer.
- The **rooms** are like virtual machines.
- The **people** are like different users or apps.
- Instead of buying a new house for each person, you are using one house in a smarter way just like virtualization lets one computer do the work of many

Need of Virtualization



Virtualization Reference Model,

- The **Virtualization Reference Model** explains how multiple virtual systems can run on a single physical system using a virtualization layer, helping in better **resource utilization**, **cost-saving**, and **isolation** between systems.
- A **Virtualization Reference Model** is a conceptual framework that explains how virtualization works in a layered manner. It shows how the **guest (virtual machine)** runs on a **host (physical machine)** through a **virtualization layer (hypervisor)**.
- This model helps in understanding how multiple virtual machines can run on a single hardware system by separating hardware and software layers.



Virtualization

- **APP**

→ The **application** running on top of the virtual machine (Guest OS).

- **Binaries/Libraries**

→ Required files and dependencies that support the application.

- **Guest OS**

→ The **Operating System** inside the virtual machine. Each VM has its own Guest OS.

- **Hypervisor**

→ Also called the **virtualization layer**, it manages virtual machines and distributes host resources to them.

- **Host OS**

→ The OS of the **physical machine**, which supports the hypervisor and manages hardware.

- **Server Hardware**

→ The **actual physical system** (CPU, RAM, storage, etc.).

Characteristics of a Virtualized Environment

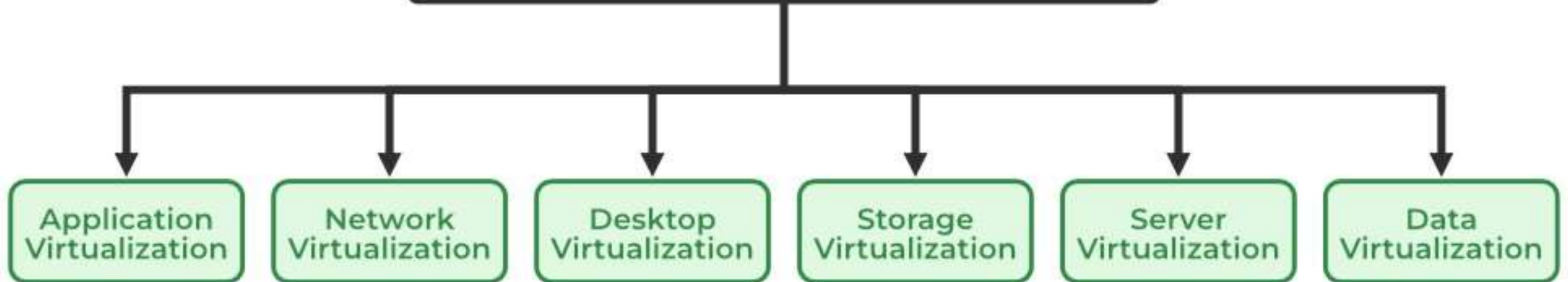
- **Increased Security** – Guest programs are isolated and filtered to block harmful operations.
- **Managed Execution** – Virtualization enables controlled execution with features like isolation and emulation.
- **Sharing** – Multiple virtual machines share the physical resources of a single host.
- **Aggregation** – Combines multiple physical hosts to act as one virtual system.
- **Emulation** – Creates a different environment to run programs not supported by the host.
- **Isolation** – Each VM runs in a separate and protected environment.
- **Portability** – Virtual machines can be easily moved across different hardware platforms.
- **Flexibility** – VMs can be created, modified, or deleted as needed without hardware changes.

Characteristics of a Virtualized Environment

- **Scalability** – Resources can be increased or decreased based on demand.
- **Centralized Management** – All virtual machines can be controlled from a single console.
- **Disaster Recovery** – VMs can be backed up and restored quickly during system failures.
- **Testing and Development** – VMs allow easy setup of test environments without extra hardware.
- **Energy Efficiency** – Reduces power usage by consolidating multiple servers into one.
- **Cost Savings** – Saves costs on hardware, power, and IT management.
- **Improved Uptime** – VMs can be migrated between hosts to reduce downtime.
- **Cloud Migration** – Virtualized systems are easier to shift to cloud platforms.

Types of Virtualization

Types of Virtualization



1.Application Virtualization

- **Definition:**

Application virtualization allows users to run applications without installing them on their local devices. The app runs on a remote server and is accessed over the internet.

- Applications are stored and executed on a central server.
- Users access the app through a web browser or thin client.
- Personal data and settings are stored on the server.
- Allows running multiple versions of the same app.
- Saves time and avoids local system compatibility issues.

Example:

Microsoft Azure - users can run apps from the cloud on any device (PC, laptop, tablet) without installing them.

2. Network Virtualization

- **Definition:**

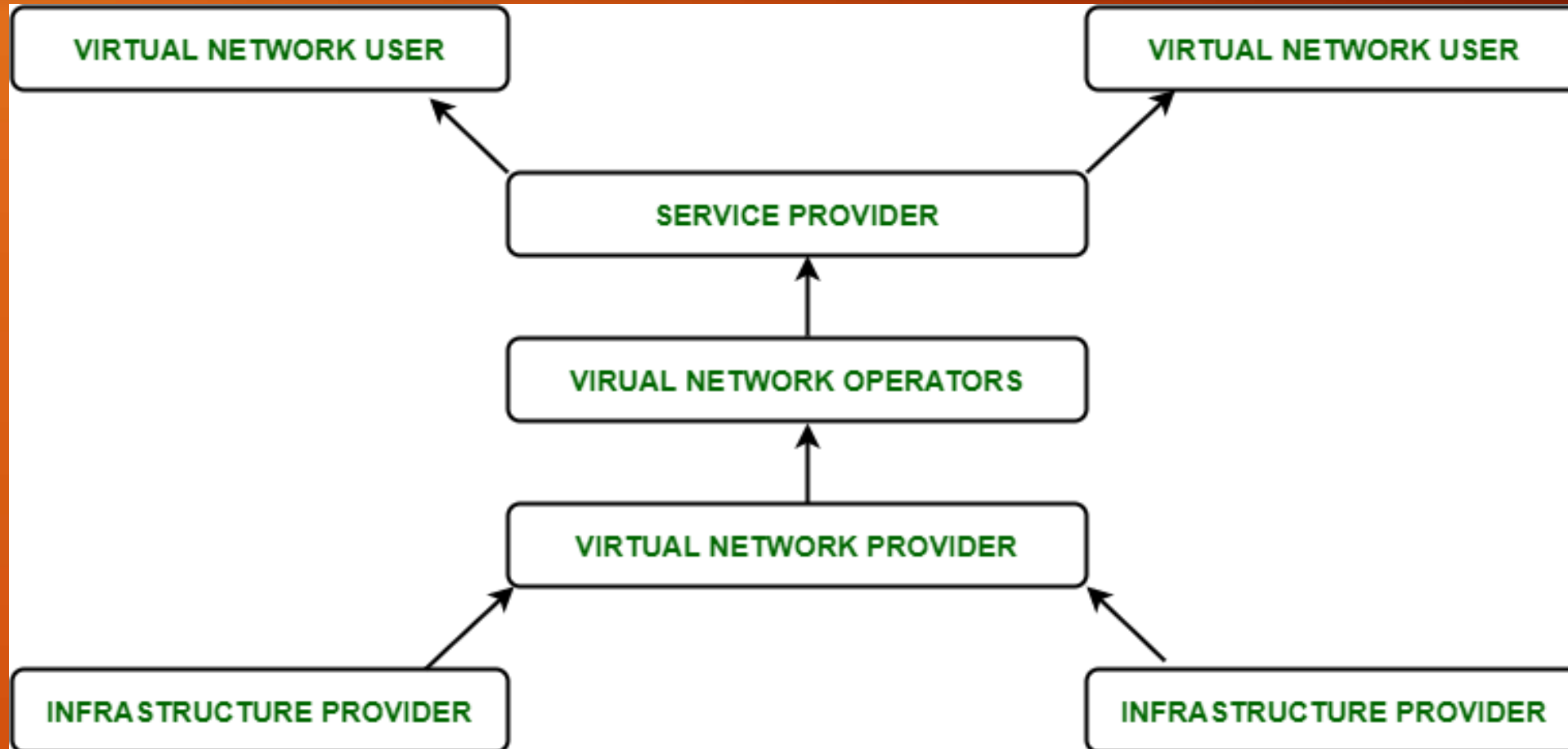
Network virtualization creates multiple virtual networks over a single physical network infrastructure.

- Virtual switches, routers, and firewalls can be created using software.
- Each virtual network works independently.
- Helps in better traffic management and security.
- Reduces dependency on physical network devices.
- Makes network changes faster and cost-effective.

- **Example:**

Google Cloud - allows companies to design and manage their network settings virtually without physical hardware.

Network Virtualization - diagram



3. Desktop Virtualization

- **Definition:**
Desktop virtualization provides virtual desktops that users can access from any device and location.
- Users get the same desktop experience on any device.
- All data and apps are stored in the cloud.
- Easy to update, manage, and secure.
- Great for remote work and team collaboration.
- Reduces hardware dependency.
- **Example:**
Amazon WorkSpaces - used by companies like GeeksforGeeks to give team members access to cloud desktops with coding tools.

4. Storage Virtualization

- **Definition:**

Storage virtualization combines storage from different servers or devices into one central pool for easier management.

- Multiple physical storage devices act like a single storage system.
- Improves storage efficiency and usage.
- Easier to manage backup and disaster recovery.
- Can continue working even if one device fails.
- Supports scalability and performance.

- **Example:**

Amazon S3 - allows businesses to store and access unlimited data from a central cloud storage.

5. Server Virtualization

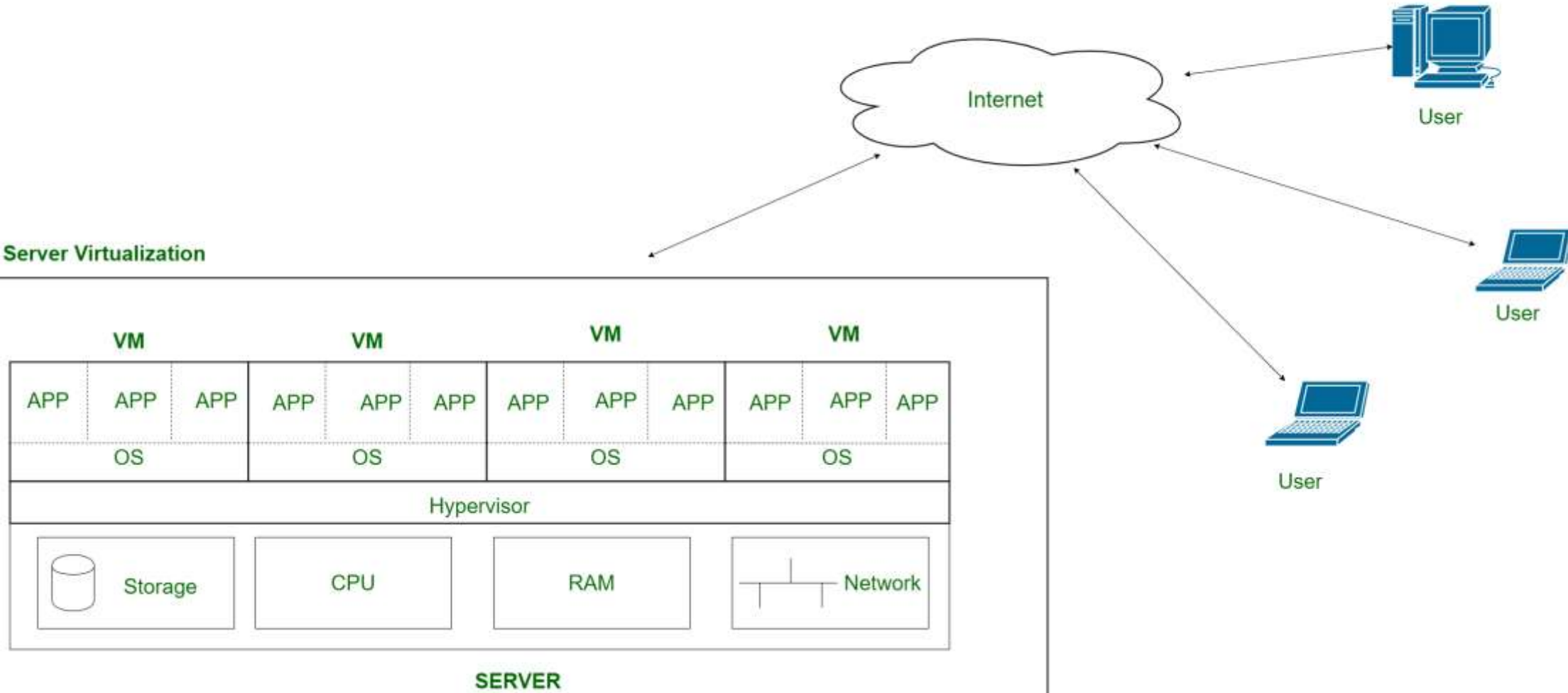
- **Definition:**

Server virtualization splits one physical server into many virtual servers (VMs), each running its own OS and applications.

- Each virtual server is independent and isolated.
- Reduces hardware cost and power consumption.
- Increases performance and uptime.
- Useful for hosting multiple services on one server.
- Easy to back up, move, or restore.
- **Example:**
Using *VMware vSphere*, a company can run a web server, database server, and file server on a single machine.

Cloud Computing

Server Virtualization



6. Data Virtualization (Not in msbte syllabus just for knowledge)

- **Definition:**

Data virtualization gives users a single view of data collected from different sources without moving or copying it.

- **Key Points:**

- No need to move or duplicate the data.
- Integrates data from different databases, systems, or clouds.
- Saves time and storage space.
- Useful in big data and analytics.
- Helps in making faster business decisions.

- **Example:**

Oracle and IBM offer data virtualization tools to combine data from different departments for real-time analysis

2.2 Differentiate Various Types of Virtualization:

Feature	Storage Virtualization	Network Virtualization	Desktop Virtualization	Application/Server Virtualization
Meaning	Combines storage from many sources	Creates multiple virtual networks	Creates virtual desktops	Runs apps/servers on VMs
Function	Acts like a single storage unit	Runs multiple virtual networks	Access desktops from any device	Allows multiple apps/servers to run on one system
Purpose	Easy data access & backup	Better network management	Remote access to desktops	Efficient server usage
Example	Amazon S3	Google Cloud Virtual Network	Amazon WorkSpaces	Microsoft Azure, VMware vSphere
Hardware Needed	Storage devices	Switches, routers (virtual)	Thin clients	Physical server
Advantage	Easy management, backup	Secure & flexible	Remote access, easy updates	Cost saving, improved performance

Type of Virtualization	Technology Example
Storage Virtualization	Amazon S3 – Cloud-based storage that combines all data in one place.
Network Virtualization	Google Cloud – Allows virtual routing, firewall, IP assignment.
Desktop Virtualization	Amazon WorkSpaces – Provides cloud desktops accessible from any device.
Application Virtualization	Microsoft Azure – Runs apps from the cloud without installing them locally.
Server Virtualization	VMware vSphere, Microsoft Hyper-V – Creates multiple virtual servers on one machine.
Data Virtualization	IBM Cloud Pak, Oracle Data Virtualization – Combines data from multiple sources.

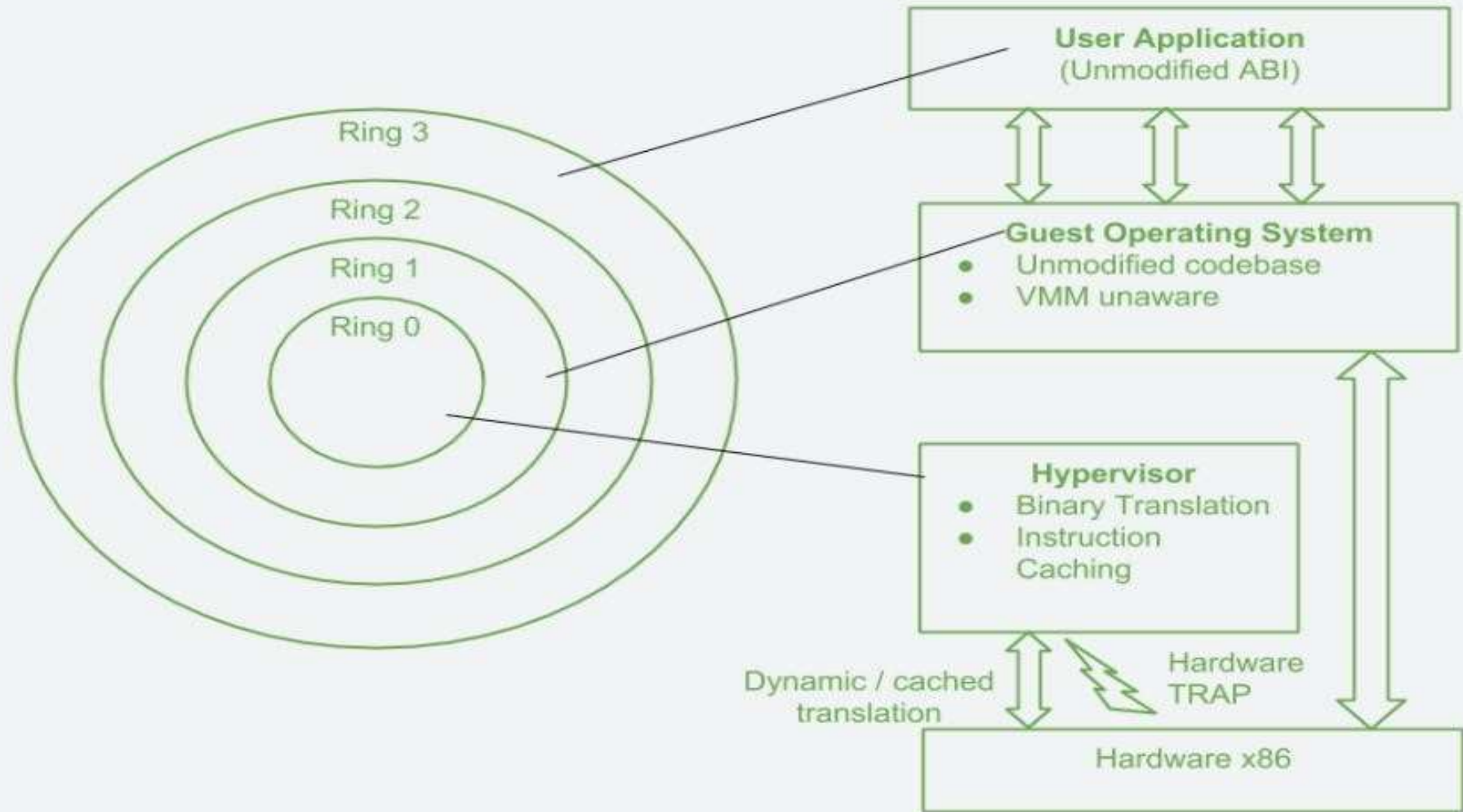
VMWare: Full Virtualization Reference Model



- Full Virtualization is a technique in which the **entire hardware is simulated** to the guest operating system.
- The **guest OS runs unmodified** as if it is directly running on physical hardware, unaware that it is in a virtualized environment.
- VMware uses Type-I (bare-metal) and Type-II (hosted) hypervisors for full virtualization. Full virtualization is achieved using a combination of: Direct execution of non-sensitive instructions Binary translation for sensitive instructions (especially on x86 architecture)
- x86 architecture does not support traditional virtualization as it doesn't trap all sensitive instructions.

CONTINUED

- VMware uses dynamic binary translation to solve this: Sensitive instructions are translated at runtime into safe instructions. These translated instructions are cached for performance. Guest OS runs in Ring 1, while actual hardware instructions run in Ring 0 (hypervisor level).
- Full virtualization in VMware allows running **unmodified guest operating systems** using **dynamic binary translation**. Although it introduces some **performance overhead**, it ensures **compatibility, security, and isolation**, especially useful for **x86 architecture**.



Full Virtualization Reference Model

Xen: Architecture and Guest Operating System Management

Introduction to Xen

- Xen is an open-source Type-I (bare-metal) hypervisor.
- It allows multiple guest operating systems to run simultaneously on a single physical machine.
- It uses a technique called paravirtualization, but also supports full virtualization
- The Xen architecture has **three main components**:
 1. Hypervisor (Xen Layer)
 2. Domain 0 (Dom0)
 3. Guest Domains (DomU)

A. Hypervisor (Xen Layer)

- Runs directly on the hardware (bare metal).
- Controls CPU, memory, and interrupt handling.
- It is a **thin layer**, lightweight, and secure.
- Provides **abstraction and isolation** for guest operating systems.

B. Domain 0 (Dom0)

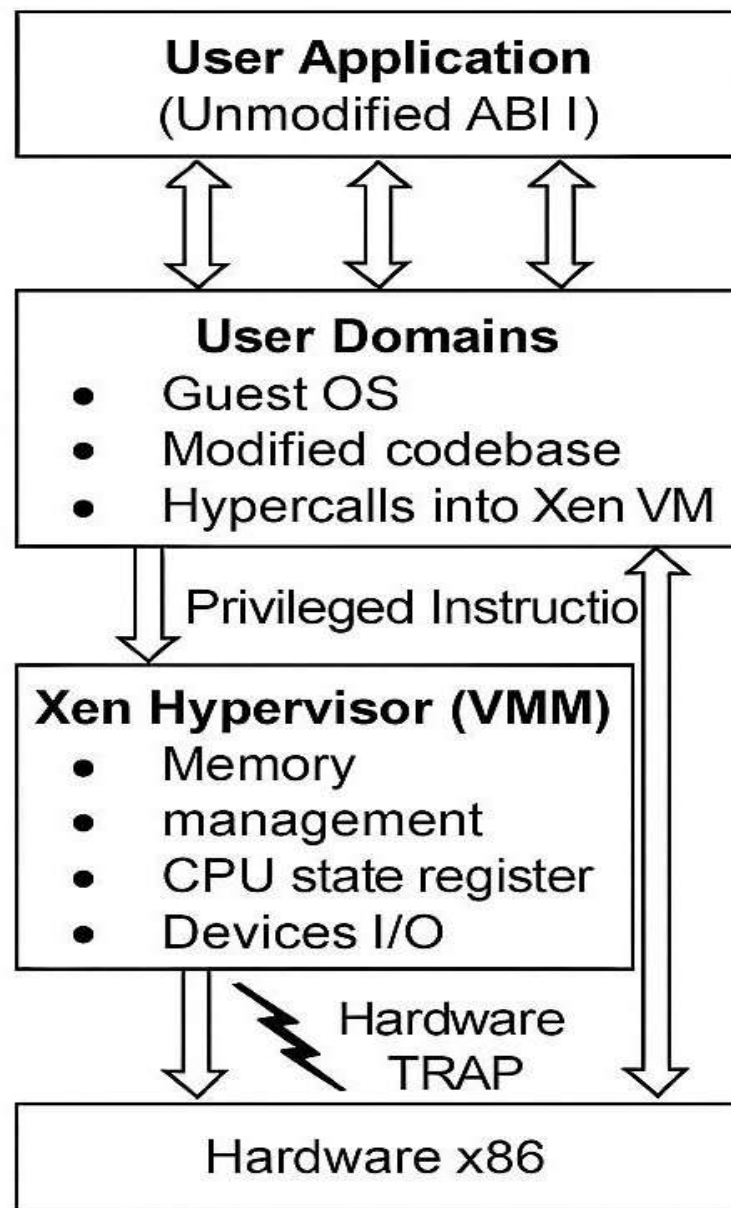
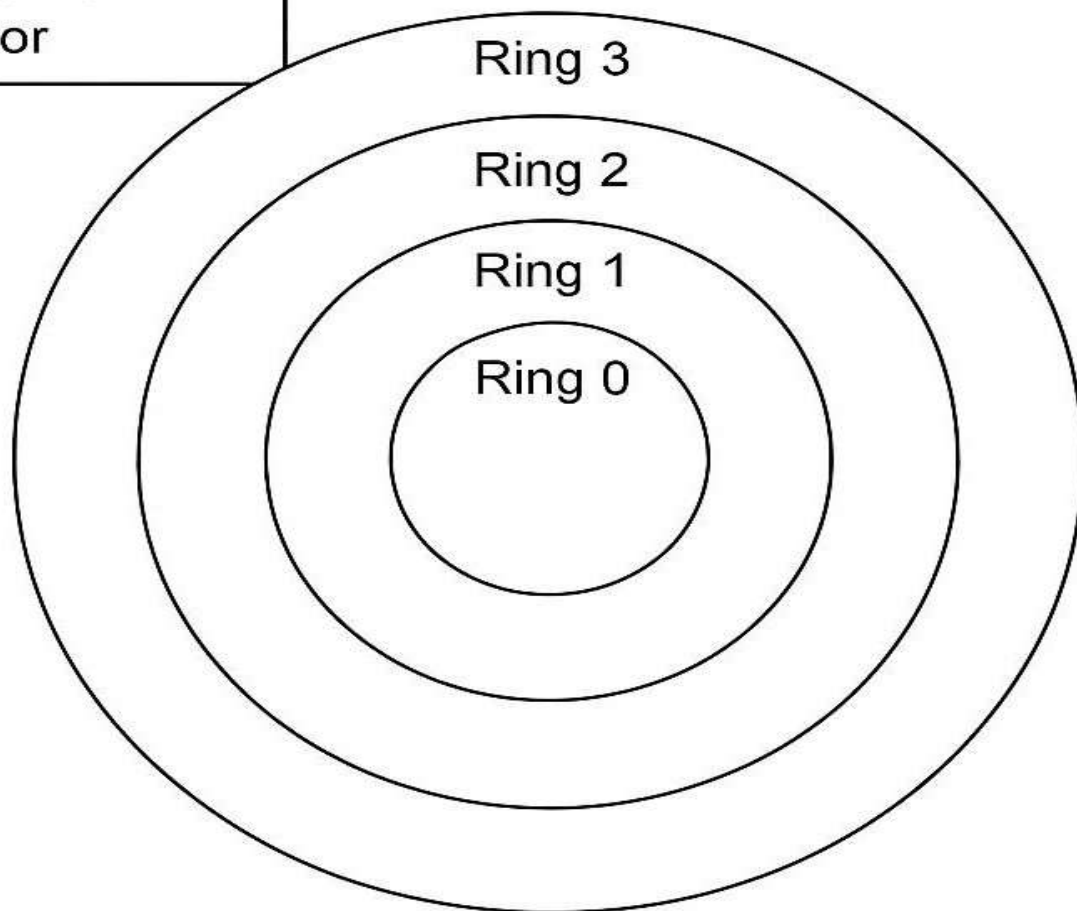
- The **first VM** started by Xen after boot.
- Has **direct access to hardware** (network, storage, etc.).
- Acts as the **management domain** and controls other VMs (DomU).
- Runs a **modified Linux kernel** with Xen management tools.

C. Guest Domains (DomU)

- These are the actual **user virtual machines**.
- Can be fully virtualized (HVM) or paravirtualized (PV).
- Do not have direct hardware access; rely on Dom0 for I/O operations.
- Used to run applications, services, or different OSes.

Management Domain

- VM Management
- HTTP Interface
- Access to Xen Hypervisor



✓ Left Side: x86 Protection Rings

- **Ring 0 to Ring 3:** Shows CPU privilege levels.
- **Xen Hypervisor:** Runs **below Ring 0**, has full control.
- **Dom0:** Privileged domain — handles **VM management, HTTP interface, access to Hypervisor.**
- **DomU:** Guest domains with limited access.

✓ Right Side: Xen Architecture

- **User Apps:** Run on top (unmodified).
- **DomU:** Guest OS (modified) uses **hypercalls** to talk to Xen.
- **Xen Hypervisor:** Manages **CPU, memory, I/O.**
- **x86 Hardware:** Executes instructions; traps on privileged calls.

Features of Xen Architecture

- **High performance** due to bare-metal hypervisor.
- **Secure isolation** between VMs.
- **Supports both PV and HVM** guests.
- **Efficient resource usage** with centralized management via Dom0.

Guest OS Management in Xen

- Dom0 manages creation, starting, stopping, and monitoring of guest OS (DomU).
- Tools like `xm`, `xl`, or `libvirt` are used for VM control.
- Dom0 allocates **CPU, memory, disk, and network** to each guest.
- Xen supports **live migration** - moving a running VM from one host to another without downtime



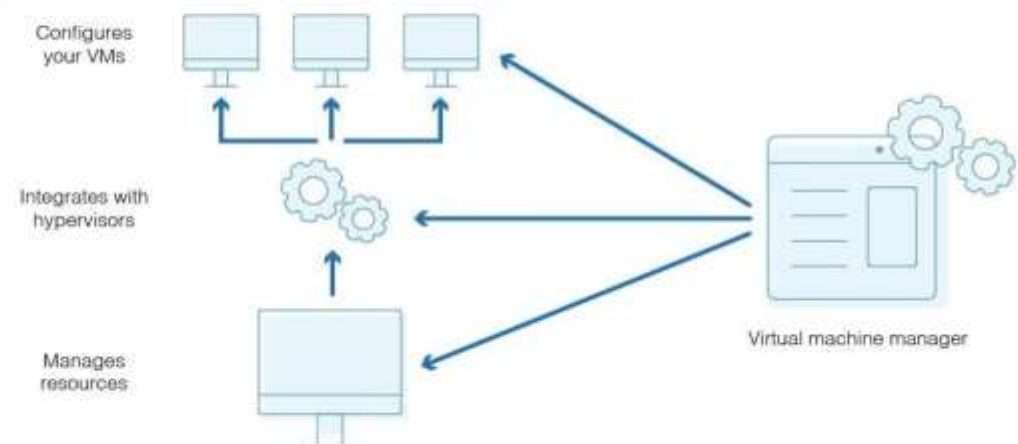
2.4 Definition and Life Cycle of Virtual Machine(VM), VM Migration: Concept and Techniques, VM Consolidation: Concepts, VM Management: Concepts

Definition of Virtual Machine (VM)

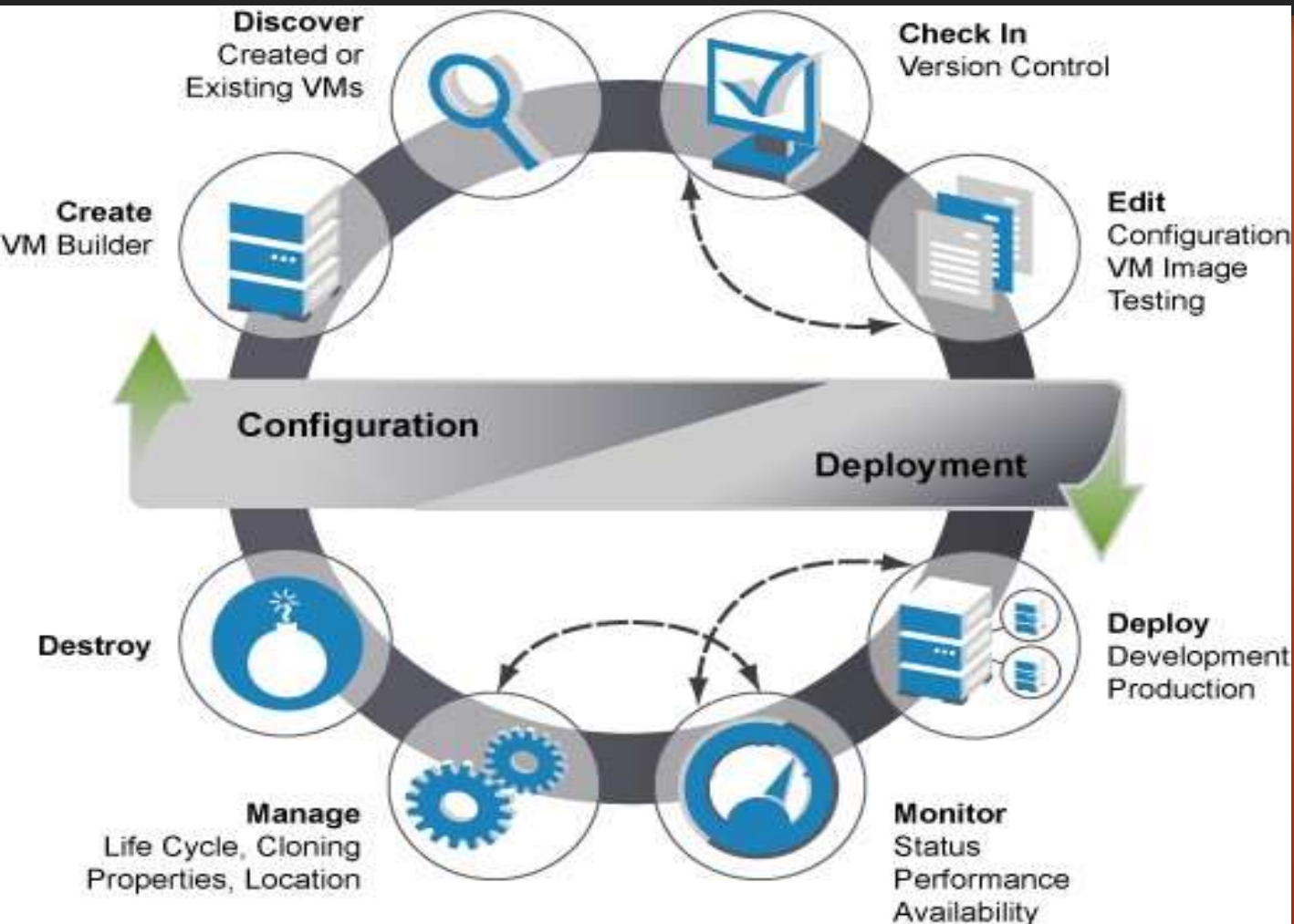
- A **Virtual Machine (VM)** is a software-based emulation of a physical computer. It runs its own OS and applications like a real computer. VMs are managed by a Hypervisor (Virtual Machine Monitor).



What Does a Virtual Machine Manager Do?



Life Cycle of a VM

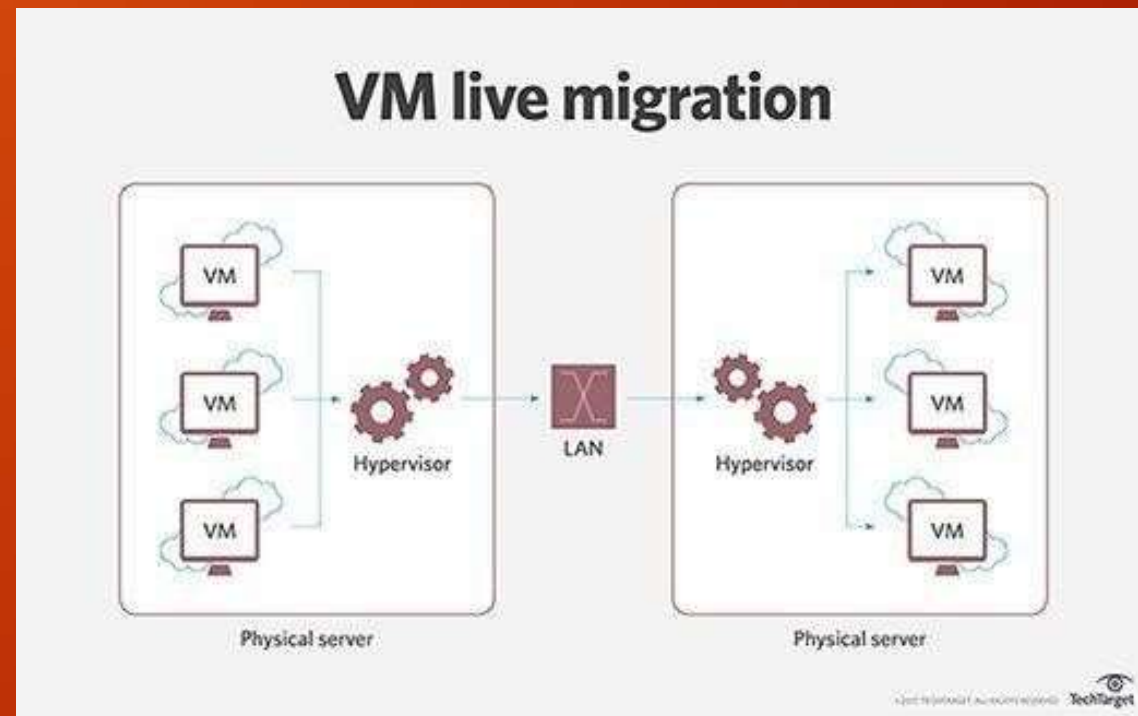


The Virtual Machine (VM) life cycle refers to the various stages a VM goes through—from creation, running, and managing to termination. It helps in tracking and managing VMs efficiently in cloud or virtualized environments.

Stage	Description
Create	VM is created with defined CPU, RAM, etc.
Deploy	OS and software are installed; VM is ready.
Run	VM is powered on and used by applications.
Pause/Save	VM is temporarily stopped or saved to disk.
Migrate	VM is moved to another host (live/cold).
Shutdown	VM is stopped normally.
Delete	VM is permanently removed from the system.

VM (Virtual Machine) migration

- VM (Virtual Machine) migration is the process of moving a running VM from one physical host to another, or from one data center to another, without disrupting the VM's availability or performance. This capability is crucial for optimizing resource utilization, balancing workloads, and ensuring high availability in virtualized environments.



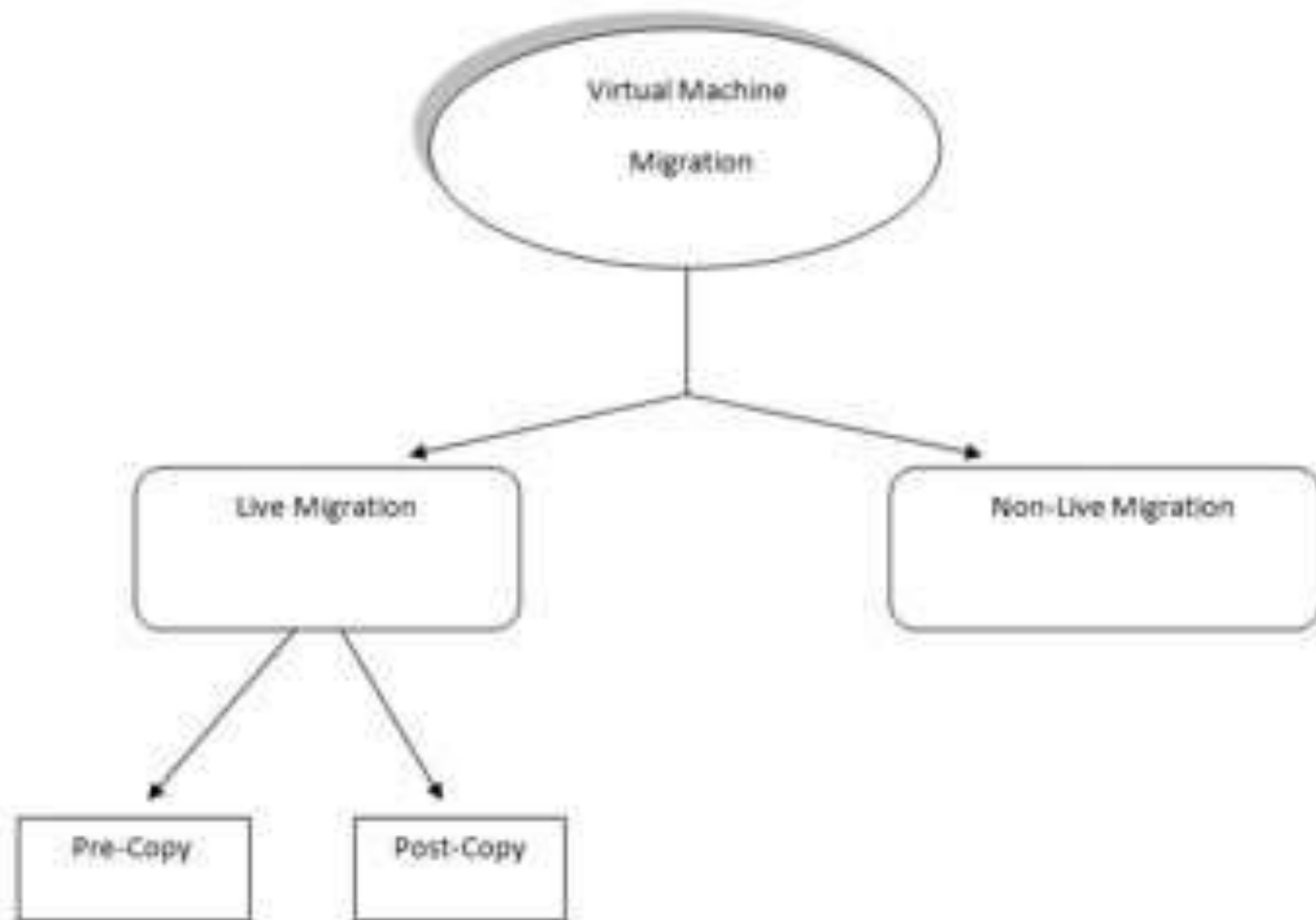
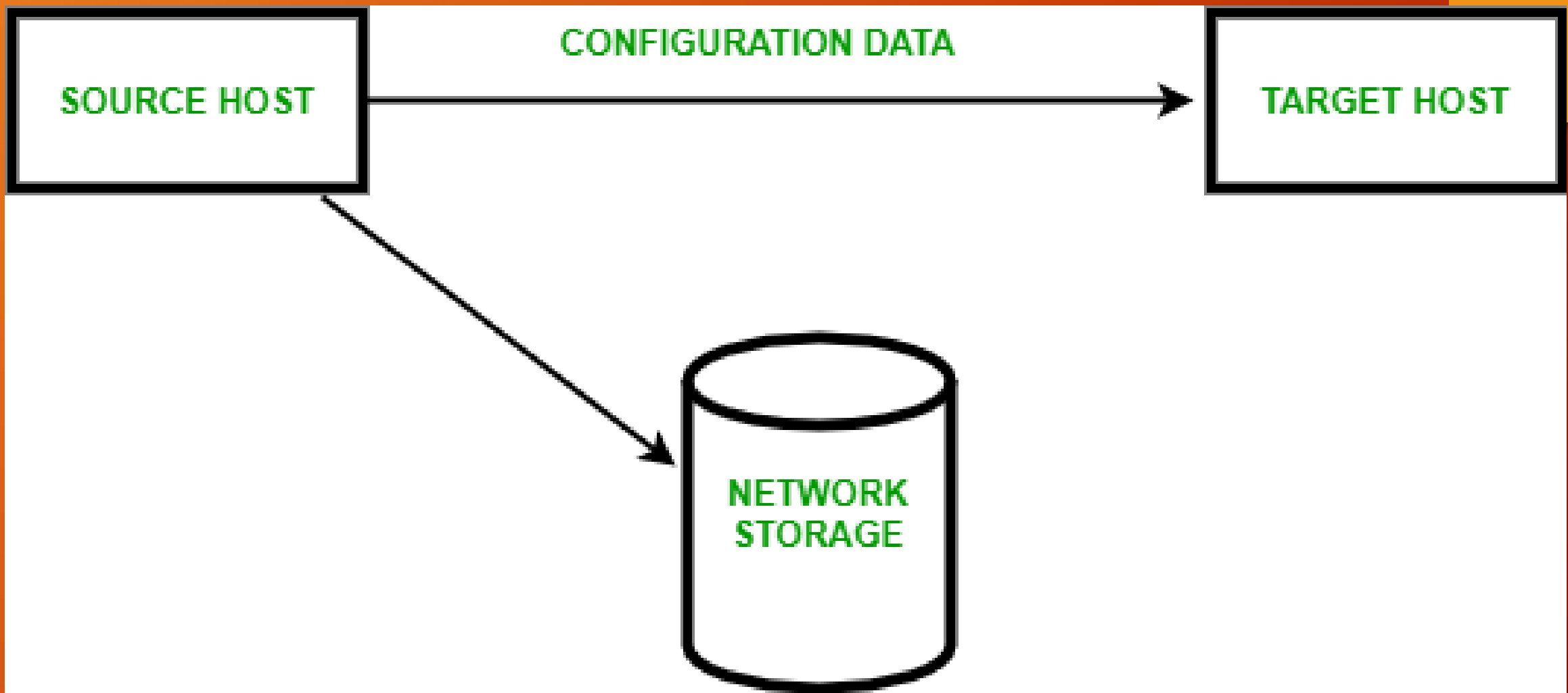


Figure 2: VM Migration

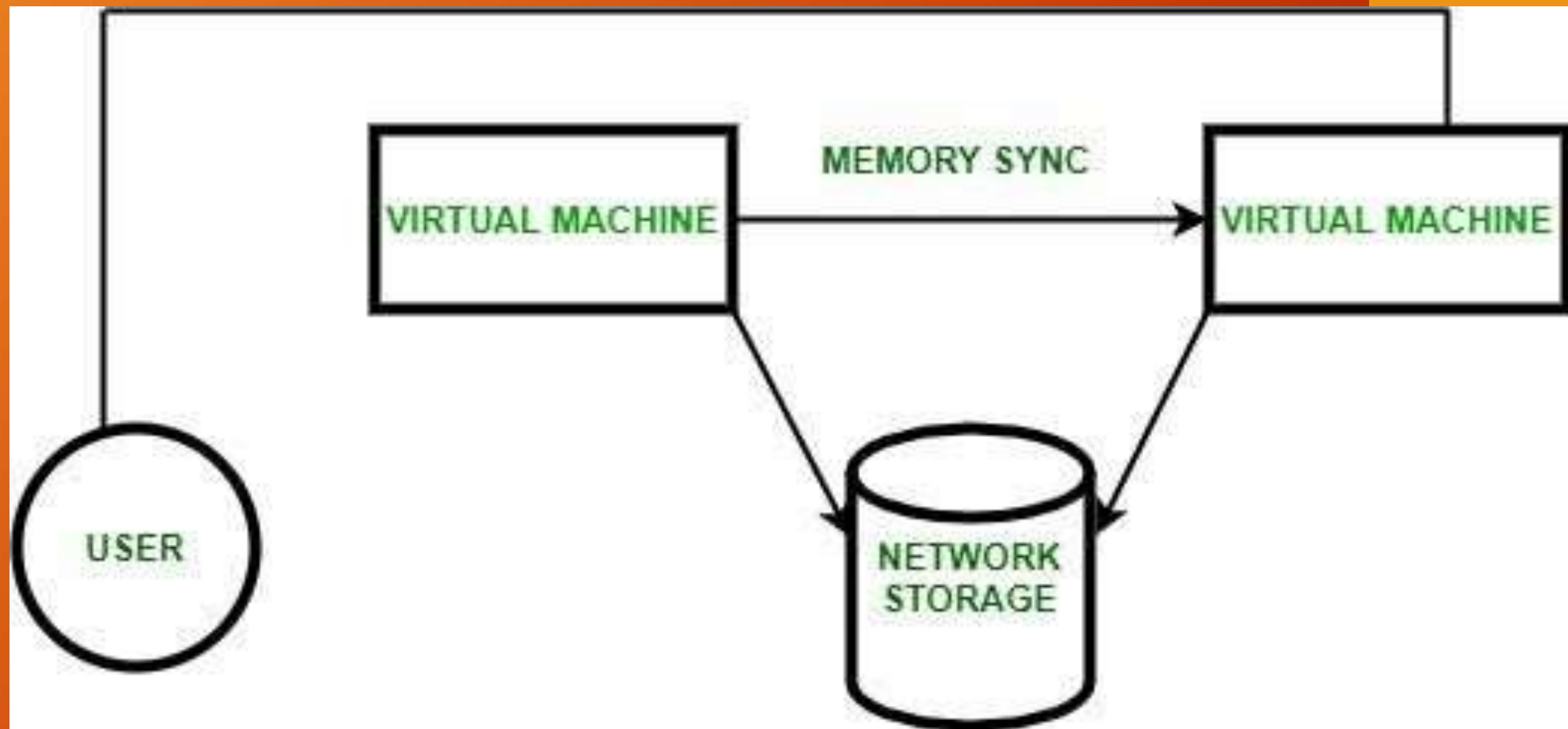
◆ Cold Migration (Non-Live Migration)

- VM is powered off before migration.
- Called non-live because the OS and applications are not running.
- No need for shared storage between source and destination hosts.
- Used for: Maintenance, upgrades, or when downtime is acceptable.
- Downtime is longer as VM must be shut down and restarted.
- Simpler process: copy files → restart VM on new host.



Hot Migration (Live Migration)

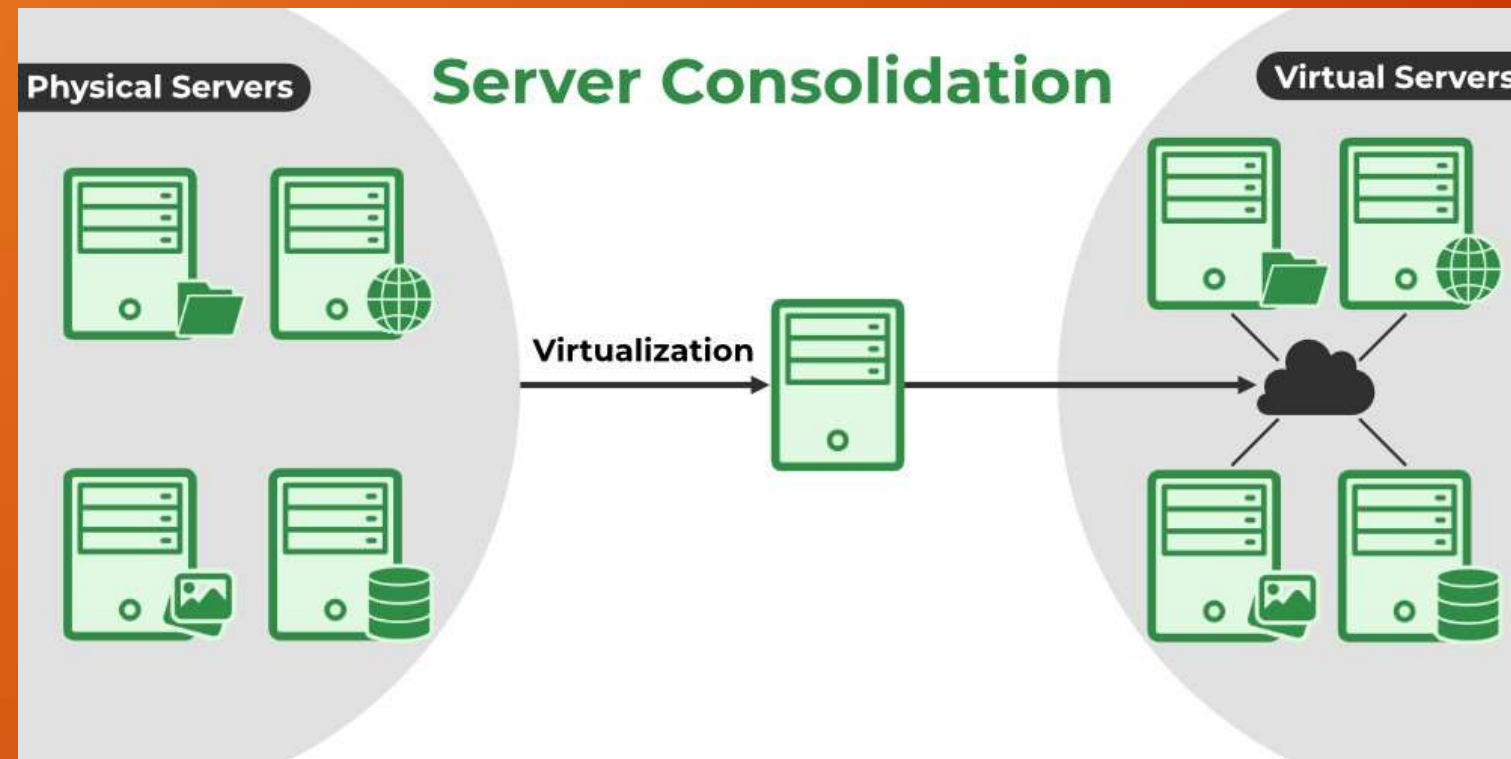
- VM is running (powered on) during the move.
- Called live migration because the VM keeps working during transfer.
- Requires shared storage and compatible CPUs.
- Used for: High-availability systems, load balancing, hardware maintenance.
- Very little or no downtime, as memory and state are copied live.
- Steps include: pre-copy memory → pause → final sync → resume on new host.



Homework

- Difference between Cold Migration (Non - Live migration) and Hot Migration (Live migration)- 6Marks

VM Consolidation



- **VM Consolidation** is the process of combining multiple virtual machines (VMs) onto fewer physical servers to improve resource usage and reduce costs.
- **Example:**
- If 10 servers are running at low usage, VMs can be moved to 4 servers, and 6 can be powered off — saving energy.

VM Consolidation: Concepts

Concept	Description
Goal	Save energy, reduce hardware usage, and cut costs.
How it works	Monitors server workloads and migrates VMs to fewer physical machines.
Helps in	Better CPU, RAM, and storage utilization.
Used with	Live migration to move VMs without downtime.
When needed	During low load periods or server maintenance .

VM Management

- VM Management refers to the processes and tools used to control, monitor, and maintain virtual machines (VMs)
- **VM Management** refers to the **control, monitoring, and maintenance** of virtual machines throughout their lifecycle
- **Benefits of Proper VM Management:**
 - Saves resources and cost
 - Increases system performance
 - Ensures uptime and reliability
 - Simplifies IT operations

2.5 Advantages and Disadvantages of Virtualization

Advantages of Virtualization

- **Cost Saving** - Reduces hardware and energy costs.
- **Better Resource Utilization** - Uses CPU, RAM, and storage efficiently.
- **Easy Backup and Recovery** - Snapshots allow quick recovery.
- **Faster Deployment** - VMs can be created quickly.
- **Isolation** - Each VM is separate from others, reducing risks.
- **Flexibility and Scalability** - VMs can be easily moved or scaled.
- **Testing Environment** - Safe space for testing software or OS.
- **Disaster Recovery** - Easy to restore from backup in case of failure.

2.5 Advantages and Disadvantages of Virtualization

Disadvantages of Virtualization

- **Initial Setup Cost** - Hypervisors and storage may cost more.
- **Performance Overhead** - VMs may run slower than physical machines.
- **Security Risks** - If hypervisor is hacked, all VMs are at risk.
- **Complex Management** - Needs skilled admins to manage VMs.
- **Limited Hardware Support** - Some hardware or software may not work in VMs.
- **Resource Contention** - Multiple VMs sharing same host can cause slowdowns.
- **Licensing Issues** - Software licensing can be more complex.

THANK YOU

